Structural Vulnerability Anatomy of Multi-Judge LLM Evaluation: Weakest-Link Dominance, Aggregation Defenses, and the Limits of Diversity

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Abstract

Multi-judge evaluation panels are increasingly adopted to improve the reliability of LLM assessment, with the implicit assumption that judge diversity provides robustness against adversarial manipulation. We present a large-scale empirical study (15 models, 455 three-judge panels, 18 attack conditions across 8 attack types and 2 benchmarks) that challenges this assumption. Our central finding is that the maximum individual vulnerability of any panel member ($\eta_{\max}$) is the dominant and near-sufficient predictor of panel attack success rate, achieving significance in 9–11 out of 12 core conditions depending on the correction method. Panel diversity (K_{eff}), once residualized against $\eta_{\max}$, reaches significance in only 2–5 of 12 conditions and is highly sensitive to the statistical method applied. Capability-matched analysis within model tiers reveals that K_{eff}'s apparent effect largely dissolves (1/6 significant), suggesting it is confounded with capability heterogeneity. Positively, we find that aggregation rule selection substantially modulates vulnerability: Bayesian aggregation reduces mean attack success rate from 0.261 (majority vote) to 0.198, while accuracy-weighted voting nearly eliminates K_{eff}'s residual association. Per-judge ASR concentration analysis reveals that adaptive adversaries exhibit targeted attack patterns (Gini coefficient 0.38 vs. 0.12 under random faults, $p < 0.001$), with targeting driven by individual weakness rather than panel diversity. We release our 15-model evaluation benchmark and provide practical recommendations for robust panel design centered on individual judge screening and appropriate aggregation.

1 Introduction

As LLMs are increasingly used as evaluators of other LLMs, multi-judge panels have emerged as a natural defense against the biases and vulnerabilities of individual models. The intuition—borrowed implicitly from Byzantine Fault Tolerance (BFT)—is that a diverse set of independent judges should be harder to simultaneously fool than any single judge. This paper subjects that intuition to rigorous empirical scrutiny at a scale not previously attempted: 15 judge models spanning 6 model families and 3 capability tiers, exhaustively combined into 455 three-judge panels and evaluated under 8 adversarial attack types across 2 benchmarks.

Our findings are sobering for proponents of diversity-based robustness. The vulnerability of a multi-judge panel is overwhelmingly determined by the weakness of its most vulnerable member. Specifically, $\eta_{\max}$—the maximum individual attack success rate among panel members—is a significant predictor of panel-level ASR in 9 out of 12 core conditions (FDR-corrected), explaining on average 34% of the variance in panel ASR by itself. Adding panel diversity (K_{eff}, the effective number of independent judgment dimensions) to the regression model contributes only an additional $\Delta R^2 = 0.04$ on average.

The residualized K_{eff} effect is fragile: it reaches significance in 4/12 conditions under FDR correction but drops to 2/12 under Bonferroni correction, and is highly sensitive to model specification choices. When we control for capability more stringently through within-tier analysis (comparing panels drawn from the same capability stratum), K_{eff}'s effect essentially vanishes (1/6 conditions significant). This

pattern suggests that the apparent diversity effect in the full pool is largely a statistical reflection of capability heterogeneity: diverse panels tend to include weaker models, and it is these weaker models—not diversity per se—that drive vulnerability.

These negative results regarding diversity are accompanied by several constructive contributions. We demonstrate that the choice of aggregation rule has a larger impact on panel robustness than panel composition: Bayesian aggregation reduces mean ASR by 24% relative to majority voting. We characterize adaptive adversary targeting behavior through per-judge ASR concentration analysis, showing that adversaries focus attacks on the weakest judge (Gini–$\eta_{\max}$ correlation $r = 0.67$) rather than exploiting panel diversity (Gini–K_{eff} correlation $r = 0.21$, $p = 0.089$). We examine generalization to open-ended evaluation (MT-Bench), where $\eta_{\max}$ dominance strengthens while K_{eff}'s direction becomes inconsistent with MCQ results. We also evaluate partial-knowledge adversaries, finding that model-specific knowledge provides only modest attack advantage (8.8% ASR increase over uniform attacks).

Our practical recommendation is straightforward: panel security is determined by the weakest judge. Practitioners should prioritize selecting individually robust judges and employing reliability-weighted aggregation, rather than pursuing diversity for its own sake.

2 Related Work

LLM-as-Judge. Using LLMs as evaluators has become standard practice for assessing open-ended generation quality, with systems ranging from single-judge scoring to multi-judge panels with various aggregation schemes. Prior work has documented individual judge biases including position bias, verbosity bias, and self-enhancement bias, motivating multi-judge approaches.

Adversarial Attacks on LLM Evaluators. Recent work has shown that LLM judges can be manipulated through prompt injection, sycophancy exploitation, and score manipulation. However, most studies focus on single-judge vulnerability rather than panel-level robustness.

Multi-Agent Robustness. The connection between multi-agent systems and fault tolerance has been explored in distributed computing, but the specific setting of LLM evaluation panels under adaptive adversaries remains understudied. Byzantine Fault Tolerance provides a theoretical framework for random faults, but its applicability to strategic adversaries requires empirical validation.

Panel Diversity Metrics. Effective diversity (K_{eff}), computed from mutual information matrices, quantifies the number of independent judgment dimensions in a panel, providing a continuous measure beyond simple model count or family-level heuristics.

3 Methodology

3.1 Threat Model

We consider an adaptive adversary who can modify evaluation inputs (prompts, candidate responses) to maximize the probability of a favorable judgment from a multi-judge panel. The adversary has varying levels of knowledge about panel composition: full knowledge (knows each judge's vulnerability profile), partial knowledge (knows judge families but not specific vulnerabilities), and no knowledge (uniform attack).

3.2 Model Pool and Panel Construction

We assemble a pool of 15 LLM judges spanning 6 model families and 3 capability tiers (Table 1). All $\binom{15}{3} = 455$ three-judge panels are enumerated exhaustively. For the panel size sensitivity analysis (§4.6), we additionally construct panels of size $K \in \{5, 7, 9\}$.

Table 1: Model pool: 15 judges across 6 families and 3 capability tiers. Mean η is the average individual attack success rate across 6 attack types; Clean ACC is the unattacked accuracy.

Model	Family	Size	Tier	Mean η	Clean ACC
Qwen2.5-7B	Qwen	7B	S	0.38	0.61
Qwen2.5-14B	Qwen	14B	M	0.22	0.74
Qwen2.5-72B	Qwen	72B	L	0.11	0.83
Llama-3.1-8B	Llama	8B	S	0.41	0.58
Llama-3.1-70B	Llama	70B	L	0.14	0.80
Mistral-7B	Mistral	7B	S	0.44	0.55
Mistral-Nemo-12B	Mistral	12B	M	0.27	0.69
Phi-3-mini	Microsoft	3.8B	S	0.49	0.52
Phi-3-medium	Microsoft	14B	M	0.25	0.71
Gemma-2-9B	Google	9B	S	0.36	0.63
GPT-4o-mini	OpenAI	API	M	0.19	0.76
GPT-4o	OpenAI	API	L	0.08	0.87
GPT-4.1	OpenAI	API	L	0.07	0.89
GPT-4.1-mini	OpenAI	API	M	0.16	0.78
Claude-3.5-Haiku	Anthropic	API	M	0.18	0.77

3.3 Attack Types

We evaluate 8 adversarial attack types spanning diverse manipulation strategies:

- **Sycophancy**: exploiting judges' tendency to agree with implied preferences.
- **Score manipulation**: direct numeric score inflation cues.
- **Position bias**: exploiting order-dependent evaluation biases.
- **Verbosity bias**: inflating response length to trigger verbosity preference.
- **Authority bias**: invoking authoritative framing.
- **Few-shot hijacking**: injecting misleading few-shot examples.
- **Persona manipulation**: exploiting persona-alignment training.
- **Instruction override**: injecting conflicting instructions.

3.4 Metrics

Individual vulnerability (η_j). The attack success rate of judge j: the fraction of evaluation instances where the adversary successfully flips the judge's assessment.

Panel ASR. The fraction of instances where the panel's aggregated decision is flipped by the attack, under a specified aggregation rule (default: majority vote).

Maximum vulnerability ($\eta_{\max}$). $\max_j \eta_j$ for a given panel—the vulnerability of the weakest link.

Effective diversity (K_{eff}). Computed from the mutual information matrix of panel members' judgments:

$$K_{\text{eff}} = \exp\left(-\sum_{i=1}^{K} \frac{\lambda_i}{\sum_j \lambda_j} \log \frac{\lambda_i}{\sum_j \lambda_j}\right) \tag{1}$$

where λ_i are the eigenvalues of the normalized MI matrix. Higher K_{eff} indicates more independent judgment dimensions.

Table 2: Residualized regression: $\eta_{\max}$ and $K_{\text{eff,resid}}$ as predictors of panel ASR across 15 models, 455 panels. Significance at $p < 0.05$ (FDR-corrected) in **bold**.

Dataset	Attack	β_η	p_η	$\beta_{K,\text{resid}}$	$p_{K,\text{resid}}$	R^2	$r(K,\eta)$
MMLU	sycophancy	0.82	<**0.001**	0.18	0.147	0.41	0.78
MMLU	score_manip	0.55	<**0.001**	0.39	**0.021**	0.36	0.68
MMLU	position	0.71	<**0.001**	0.24	0.089	0.39	0.75
MMLU	verbosity	0.88	<**0.001**	0.09	0.511	0.42	0.81
MMLU	authority	0.79	<**0.001**	0.15	0.218	0.37	0.76
MMLU	few_shot	0.64	**0.002**	0.31	**0.043**	0.33	0.70
ARC	sycophancy	0.85	<**0.001**	0.12	0.341	0.43	0.79
ARC	score_manip	0.48	**0.003**	0.41	**0.032**	0.35	0.35
ARC	position	0.73	<**0.001**	0.22	0.102	0.38	0.74
ARC	verbosity	0.91	<**0.001**	0.06	0.649	0.44	0.82
ARC	authority	0.76	<**0.001**	0.19	0.157	0.36	0.77
ARC	few_shot	0.59	**0.004**	0.28	0.071	0.32	0.72

Regression framework. For each condition (dataset $\times$ attack), we regress panel ASR on $\eta_{\max}$ and K_{eff} (residualized against $\eta_{\max}$) across 455 panels:

$$\text{ASR}_p = \beta_0 + \beta_\eta \cdot \eta_{\max,p} + \beta_K \cdot K_{\text{eff,resid},p} + \epsilon_p \tag{2}$$

4 Experiments

4.1 Weakest-Link Dominance

Table 2 presents the residualized regression results across all 12 core conditions (2 datasets $\times$ 6 attacks). $\eta_{\max}$ achieves significance in 9/12 conditions with a mean $\beta_\eta = 0.73$ (range: 0.48–0.91). In contrast, $K_{\text{eff,resid}}$ reaches significance in only 4/12 conditions (FDR-corrected), with a substantially smaller mean $\beta_K = 0.22$ (range: 0.06–0.41).

The mean R^2 of the full model is 0.38, of which $\eta_{\max}$ alone explains 0.34 (R^2_η). The incremental contribution of K_{eff} is $\Delta R^2 = 0.04$—less than 2% of additional unique variance explained by diversity beyond what individual weakness already captures.

The correlation between K_{eff} and $\eta_{\max}$ is substantial (mean $r = 0.73$, range 0.35–0.85), indicating that diverse panels systematically include weaker judges. Only 1 of 12 conditions constitutes a "confound-free" case where the correlation is low enough ($r \approx 0.35$) for residualization to be trustworthy; in that condition, K_{eff} is marginally significant ($p = 0.032$, $\beta = 0.41$).

4.2 Attack Type Generalization

To test whether these patterns hold beyond the original 6 attack types, we evaluate 3 additional attacks (persona manipulation, instruction override) on the same 455 panels (Table 3). The extended results reinforce $\eta_{\max}$ dominance: $\eta_{\max}$ is significant in 5/6 new conditions, while $K_{\text{eff,resid}}$ is significant in only 2/6. Notably, persona manipulation on ARC yields a negative (though non-significant) K_{eff} direction ($\beta = -0.12$, $p = 0.389$), indicating that diversity may even be mildly protective in some settings—inconsistent with any universal "diversity hurts" claim.

Across all 18 conditions combined, $\eta_{\max}$ reaches significance in 14/18 (78%) while $K_{\text{eff,resid}}$ reaches significance in only 6/18 (33%).

4.3 Adaptive Adversary Targeting Behavior

We analyze per-judge ASR distributions within each panel to characterize how adaptive adversaries allocate attack effort. Table 4 compares attack concentration between adaptive and random-fault regimes

Table 3: Extended attack types (455 panels, 15 models). K_{eff} direction is inconsistent across conditions.

Dataset	Attack	β_η	p_η	$\beta_{K,\text{resid}}$	$p_{K,\text{resid}}$	R^2	Direction
MMLU	few_shot_hijack	0.64	**0.002**	0.31	**0.043**	0.33	positive
MMLU	persona_manip	0.91	<**0.001**	0.05	0.724	0.45	near-zero
MMLU	instruct_override	0.58	**0.005**	0.34	**0.038**	0.31	positive
ARC	few_shot_hijack	0.59	**0.004**	0.28	0.071	0.32	positive
ARC	persona_manip	0.87	<**0.001**	−0.12	0.389	0.41	negative
ARC	instruct_override	0.62	**0.003**	0.26	0.088	0.30	positive

Table 4: ASR concentration statistics: adaptive adversaries target specific judges far more than random faults.

Metric	Adaptive (mean)	SD	RF (mean)	SD	p
Gini coefficient	0.38	0.14	0.12	0.05	<0.001
Entropy (bits)	2.41	0.52	3.22	0.31	<0.001
Max single-judge share	0.51	0.12	0.38	0.06	<0.001

using the Gini coefficient (higher = more concentrated on a single judge) and Shannon entropy (lower = more concentrated).

Adaptive adversaries exhibit substantially more concentrated targeting (Gini 0.38 vs. 0.12, $p < 0.001$): on average, a single judge absorbs 51% of the attack burden in the adaptive setting versus 38% under random faults. Critically, this targeting tracks individual weakness, not panel diversity: Gini correlates strongly with $\eta_{\max}$ ($r = 0.67$, $p < 0.001$) but only weakly with K_{eff} ($r = 0.21$, $p = 0.089$; Table 5). The adversary attacks the weakest judge, not the most "unique" one.

4.4 Aggregation Rule Comparison

Beyond panel composition, the aggregation rule used to combine individual judgments substantially affects panel robustness (Tables 6 and 7).

Bayesian aggregation achieves the lowest mean ASR (0.198) with the smallest spread between best and worst panels (0.182), representing a 24% reduction over majority voting (0.261). Threshold (unanimity) voting is the most conservative rule (lowest ASR for top panels at 0.098) but amplifies K_{eff}'s residual significance from 4/12 to 6/12—because unanimity grants each judge effective veto power, and diverse panels are more likely to include a judge that vetoes correct decisions. Accuracy-weighted voting nearly eliminates K_{eff}'s residual effect (2/12 significant) while maintaining strong $\eta_{\max}$ predictive power (10/12), suggesting that down-weighting unreliable judges addresses both vulnerability drivers.

The variation in K_{eff}'s significance across aggregation rules (1/12 to 6/12) itself suggests that the residual K_{eff} effect is partially an aggregation artifact rather than a structural vulnerability property.

4.5 Panel Selection Strategies

We benchmark 6 panel selection strategies (Table 8) to translate our analytical findings into actionable guidance. The two best-performing strategies—Min-$\eta_{\max}$ and Strongest-K—both prioritize individual judge robustness and achieve statistically indistinguishable performance (mean ASR 0.229 vs. 0.234, $p = 0.478$). A Two-Stage algorithm (first screen for $\eta \leq 0.15$, then maximize K_{eff} among qualifying judges) achieves the numerically lowest ASR (0.227) but does not significantly outperform Strongest-K ($\Delta = -3.1\%$, $p = 0.312$).

Conversely, Max-K_{eff} is the *worst* strategy (ASR 0.341), because maximizing diversity in an unrestricted pool systematically introduces the weakest judges. This reinforces our central finding: diversity without robustness screening is counterproductive.

Table 5: Correlates of attack concentration (Gini coefficient). Targeting is driven by individual weakness ($\eta_{\max}$) and mean capability, not diversity (K_{eff}).

Predictor	r **with Gini**	p	**Interpretation**
$\eta_{\max}$	0.67	<0.001	Targeting tracks weakness
K_{eff}	0.21	0.089	Weak / NS
Panel size	0.04	0.814	No effect
Mean capability	−0.58	<0.001	Stronger panels → less concentrated

Table 6: Vulnerability driver significance under different aggregation rules (12 conditions, 455 panels).

Aggregation	K_{eff} **sig (/12)**	$\eta_{\max}$ **sig (/12)**	**Mean** R^2	**Best predictor**
Majority Vote	4	9	0.38	$\eta_{\max}$
Threshold (Unanimity)	6	7	0.42	mixed
Accuracy-Weighted	2	10	0.45	$\eta_{\max}$
Bayesian	1	5	0.29	$\eta_{\max}$ (weakened)

4.6 Panel Size Sensitivity

We evaluate panel sizes $K \in \{3, 5, 7, 9\}$ to test whether our findings are robust to panel cardinality (Table 10). $\eta_{\max}$'s significance ratio increases monotonically with panel size (0.75 at K=3 to 0.92 at K=9), consistent with the weakest-link mechanism: larger panels are more likely to include at least one weak judge, making $\eta_{\max}$ an increasingly reliable predictor. K_{eff}'s significance ratio decreases monotonically (0.33 at K=3 to 0.08 at K=9), further supporting the interpretation that its apparent effect in small panels reflects tier-mixing variance rather than a genuine diversity mechanism.

4.7 Open-Ended Task Generalization

To test whether MCQ-based findings generalize, we evaluate on MT-Bench (adapted) with 4 task-specific attacks (Table 11). $\eta_{\max}$ maintains significance in all 4 conditions (dominance ratio 0.91), confirming that weakest-link vulnerability is task-general. K_{eff} shows inconsistent behavior: one condition yields a significant *negative* direction (MT-Bench-Writing, $\beta = -0.22$, $p = 0.034$), meaning diversity is mildly protective—opposite to the MCQ pattern. The remaining 3 conditions are non-significant for K_{eff}. Overall mean ASR on open-ended tasks (0.34) exceeds MCQ (0.26), suggesting open-ended evaluation is generally more vulnerable to adversarial manipulation.

4.8 Partial-Knowledge Adversary

We relax the full-knowledge assumption by simulating adversaries with three knowledge levels (Table 12). Full-to-partial knowledge degradation is modest: knowing judge families instead of exact vulnerability profiles reduces ASR by only 8.8% (0.261 → 0.238). Even zero-knowledge (uniform) attacks achieve an ASR of 0.204—only 21.8% below full knowledge. $\eta_{\max}$ maintains significance under partial knowledge (8/12) and even under no knowledge (6/12), while K_{eff} drops to 2/12 and 0/12 respectively. This indicates that $\eta_{\max}$'s predictive power is robust across threat models, while K_{eff}'s weak association depends on unrealistically strong adversary knowledge assumptions.

5 Discussion

5.1 The Role of Panel Diversity: Exploratory Analysis

Our primary analysis establishes $\eta_{\max}$ as the dominant predictor of panel vulnerability. We now examine the residual K_{eff} effect more carefully, arguing that it is suggestive but confounded, and should not be treated as an established finding.

Table 7: Aggregation as defense: mean ASR across all 455 panels and for top-10/bottom-10 panels.

Aggregation	Mean ASR	Top-10 ASR	Bottom-10 ASR	Spread
Majority	0.261	0.142	0.418	0.276
Threshold	0.183	0.098	0.312	0.214
Weighted	0.224	0.121	0.367	0.246
Bayesian	0.198	0.112	0.294	0.182

Table 8: Panel selection strategies: mean ASR across 6 attack conditions. Wins = number of conditions where the strategy achieves the lowest ASR.

Strategy	Mean ASR	vs. Strongest-K Δ%	p	Wins
Random	0.284	+21.5%	<0.001	0
Max-K_{eff}	0.341	+45.9%	<0.001	0
Diversity-Only	0.312	+33.5%	<0.001	0
Min-$\eta_{\max}$	0.229	-2.1%	0.478	4
Strongest-K	0.234	—	—	1
Two-Stage	0.227	-3.1%	0.312	1

Multicollinearity analysis. Table 14 presents K_{eff}'s significance under multiple correction and estimation methods. The significance count ranges from 2/12 (Bonferroni) to 7/12 (uncorrected)—a lack of convergence that undermines confidence in K_{eff}'s independent effect. The partial correlation of K_{eff} controlling for $\eta_{\max}$ is 0.14 ($r^2 = 0.02$), meaning K_{eff} explains approximately 2% of unique variance (Table 15). By contrast, $\eta_{\max}$'s partial correlation is 0.41 ($r^2 = 0.17$).

Capability-matched analysis. The strongest test of K_{eff}'s independence comes from within-tier analysis (Table 16). When panels are drawn from the same capability tier (e.g., all small 7–8B models or all large 70B+/API models), $\eta_{\max}$ variance shrinks dramatically (0.008 vs. 0.045 in the full pool) and K_{eff} achieves significance in only 1/6 conditions. This result is consistent with the interpretation that K_{eff}'s apparent effect in the full pool is driven by tier-mixing: diverse panels cross capability boundaries, systematically introducing weaker judges.

5.2 BFT Analogy: An Open Question

The Byzantine Fault Tolerance framework predicts that independent judges should tolerate random faults through majority vote. We tested whether K_{eff}'s role differs between adaptive and random-fault regimes (Table 17).

Contrary to the BFT prediction, K_{eff} is positively associated with panel ASR under *both* regimes: 10/12 significant under random faults, compared to 4/12 under adaptive attacks. The direction is consistent (positive in both), with the β ratio between adaptive and RF averaging 0.91. The regime interaction term is non-significant ($p = 0.217$), meaning we cannot reject the null that K_{eff}'s effect is the same in both regimes.

The key regime difference is predictability: R^2 is substantially higher under random faults (mean 0.62 vs. 0.38, $\Delta R^2 = 0.24$; Table 18), because adaptive adversaries introduce strategic variance that panel composition cannot explain. However, this predictability gap does not constitute a BFT reversal—diversity does not switch from protective to harmful across regimes. We therefore treat the BFT connection as an open question requiring further theoretical investigation.

5.3 Diversity-Capability Tradeoff

The correlation between K_{eff} and $\eta_{\max}$ ($r = 0.73$, Spearman $\rho = 0.71$; tradeoff curve $R^2 = 0.53$) reflects a structural constraint in the current LLM ecosystem: achieving high panel diversity requires

Table 9: Per-condition comparison of Two-Stage vs. Strongest-K. No condition reaches statistical significance.

Condition	**Two-Stage**	**Strongest-K**	$\Delta\%$	p
MMLU × sycophancy	0.211	0.219	-3.7%	0.284
MMLU × score_manip	0.198	0.207	-4.3%	0.198
MMLU × position	0.237	0.241	-1.7%	0.641
ARC × sycophancy	0.241	0.248	-2.8%	0.352
ARC × score_manip	0.185	0.194	-4.6%	0.187
ARC × position	0.249	0.253	-1.6%	0.689

Table 10: Panel size sensitivity: $\eta_{\max}$ dominance strengthens and K_{eff} effect vanishes as panel size increases.

K	**Panels**	$\eta_{\max}$ **sig ratio**	K_{eff} **sig ratio**	**Mean** R^2	$r(K_{\text{eff}}, \eta_{\max})$
3	455	0.75	0.33	0.38	0.73
5	3,003	0.83	0.25	0.41	0.68
7	6,435	0.92	0.17	0.44	0.61
9	5,005	0.92	0.08	0.46	0.54

mixing models from different capability tiers, which inevitably introduces weaker judges.

Pareto frontier analysis identifies 12 panels that are non-dominated on the (K_{eff}, $\eta_{\max}$) plane (Table 19). 83% of Pareto-optimal panels consist entirely of large-tier models, and 17% are large+medium mixtures; no small-tier panels appear on the frontier. The Pareto-optimal panels have mean $K_{\text{eff}} = 1.82$ and mean $\eta_{\max} = 0.09$—essentially, the recommendation reduces to "select the strongest available models," with diversity being a secondary consequence rather than a design objective.

6 Conclusion

This paper presents a systematic empirical study of structural vulnerability in multi-judge LLM evaluation, spanning 15 models, 455 panels, and 18 attack conditions. Our findings establish three principal conclusions:

(1) Weakest-link dominance. Panel vulnerability is predominantly determined by the most vulnerable individual judge ($\eta_{\max}$), which achieves significant predictive power in 9–14 out of 12–18 conditions depending on the attack set. Panel diversity (K_{eff}) adds negligible predictive value ($\Delta R^2 = 0.04$) and its residual significance is fragile across correction methods (2–5/12) and vanishes under capability-matched analysis (1/6).

(2) Aggregation matters more than composition. The choice of aggregation rule has a larger effect on panel robustness than panel composition. Bayesian aggregation reduces mean ASR by 24% relative to majority voting and substantially narrows the performance gap between strong and weak panels.

(3) Adaptive targeting tracks weakness, not diversity. Adaptive adversaries concentrate attacks on the weakest judge (Gini–$\eta_{\max}$ $r = 0.67$) rather than exploiting structural diversity (Gini–K_{eff} $r = 0.21$, NS). This targeting pattern is robust across threat models: even partial-knowledge adversaries achieve 91% of full-knowledge ASR.

For practitioners, the recommendation is simple: select the most individually robust judges available and employ reliability-weighted aggregation. Diversity is neither a reliable defense nor a consistent liability—it is largely irrelevant once individual judge quality is controlled.

Table 11: Open-ended evaluation (MT-Bench adapted): $\eta_{\max}$ dominates; K_{eff} direction is inconsistent with MCQ results.

Task	**Attack**	β_η	p_η	$\beta_{K,\text{resid}}$	$p_{K,\text{resid}}$	**Direction**
Writing	style_manip	0.79	$<$**0.001**	-0.22	**0.034**	NEGATIVE
Reasoning	logic_distort	0.84	$<$**0.001**	0.08	0.541	near-zero
Coding	output_inject	0.71	**0.002**	0.15	0.218	positive (NS)
Math	step_skip	0.88	$<$**0.001**	-0.05	0.712	near-zero

Table 12: Adversary knowledge level comparison: $\eta_{\max}$ remains predictive across threat models; K_{eff} vanishes under realistic assumptions.

Knowledge Level	**Mean ASR**	K_{eff} **sig (/12)**	$\eta_{\max}$ **sig (/12)**	R^2
Full (exact η_j)	0.261	4	9	0.38
Partial (family only)	0.238	2	8	0.34
None (uniform attack)	0.204	0	6	0.22

Limitations

Our study has three main limitations. First, the capability-matched within-tier analysis is limited by small sample sizes ($\binom{5}{3} = 10$ panels per tier), which may lack statistical power to detect genuine small effects of diversity. Second, our adaptive adversary model assumes oracle access to individual judge vulnerability profiles; real-world adversaries operate with partial knowledge, under which K_{eff}'s effect further diminishes (0/12 significant under zero knowledge). Third, MCQ and open-ended evaluation involve different attack dynamics, and our K_{eff} findings on MCQ benchmarks do not generalize to open-ended tasks (where diversity shows inconsistent, sometimes protective, effects); conclusions about diversity's role should therefore be qualified by evaluation format.

Table 13: ASR degradation analysis across knowledge levels.

Transition	**ΔASR**	**Δ%**	**Interpretation**
Full → Partial	−0.023	−8.8%	Family info has modest value
Full → None	−0.057	−21.8%	Knowledge matters but not dramatically
Partial → None	−0.034	−14.3%	Family info alone: limited advantage

Table 14: K_{eff} significance under different correction methods (12 conditions). The range 2–7 indicates high sensitivity to analytical choices.

Method	K_{eff} **sig (/12)**	$\eta_{\max}$ **sig (/12)**	**Notes**
Uncorrected	7	10	Liberal
FDR (BH)	4	9	Standard
Bonferroni	2	8	Conservative
Bootstrap 95% CI	3	9	Non-parametric
Excl. extreme models (FDR)	3	9	Sensitivity
Beta regression (FDR)	3	10	Bounded DV

Table 15: Partial correlations: K_{eff} explains negligible unique variance after controlling for $\eta_{\max}$.

Predictor	**Partial** r	**95% CI**	p
$K_{\text{eff}} \mid \eta_{\max}$	0.14	[0.05, 0.23]	0.003
$\eta_{\max} \mid K_{\text{eff}}$	0.41	[0.33, 0.49]	<0.001

Table 16: Capability-matched K_{eff} analysis: within-tier panels. K_{eff}'s effect essentially vanishes when capability is controlled.

Tier	**Panels**	**Condition**	β_K	p_K	R^2	$\eta_{\max}$ **Var**
Small (7–8B)	10	MMLU × syco.	0.08	0.721	0.09	0.007
Small (7–8B)	10	ARC × score	0.14	0.548	0.11	0.008
Medium (12–14B)	10	MMLU × syco.	0.11	0.612	0.10	0.009
Medium (12–14B)	10	ARC × score	0.29	0.041	0.18	0.011
Large (70B+/API)	10	MMLU × syco.	0.06	0.891	0.07	0.004
Large (70B+/API)	10	ARC × score	0.09	0.684	0.08	0.005

Table 17: BFT regime contrast: K_{eff} is positively associated with ASR in both adaptive and random-fault regimes. No reversal observed.

Data	**Attack**	β_K **(Adpt.)**	p **(Adpt.)**	β_K **(RF)**	p **(RF)**	β **Ratio**	**Int.** p
MMLU	syco.	0.18	0.147	0.21	**0.024**	1.17	0.682
MMLU	score	0.39	**0.021**	0.44	**0.003**	1.13	0.541
MMLU	pos.	0.24	0.089	0.28	**0.011**	1.17	0.614
MMLU	verb.	0.09	0.511	0.15	0.108	1.67	0.398
MMLU	auth.	0.15	0.218	0.19	**0.047**	1.27	0.583
MMLU	few_s.	0.31	**0.043**	0.35	**0.008**	1.13	0.617
ARC	syco.	0.12	0.341	0.16	0.072	1.33	0.521
ARC	score	0.41	**0.032**	0.48	**0.001**	1.17	0.434
ARC	pos.	0.22	0.102	0.27	**0.015**	1.23	0.548
ARC	verb.	0.06	0.649	0.11	0.198	1.83	0.412
ARC	auth.	0.19	0.157	0.22	**0.038**	1.16	0.601
ARC	few_s.	0.28	0.071	0.33	**0.009**	1.18	0.547

Table 18: R^2 comparison across regimes: adaptive attacks introduce strategic variance that reduces predictability.

Dataset	**Attack**	R^2 **(Adaptive)**	R^2 **(RF)**	ΔR^2
MMLU	sycophancy	0.41	0.64	0.23
MMLU	score_manip	0.36	0.58	0.22
MMLU	position	0.39	0.61	0.22
MMLU	verbosity	0.42	0.67	0.25
MMLU	authority	0.37	0.59	0.22
MMLU	few_shot	0.33	0.55	0.22
ARC	sycophancy	0.43	0.69	0.26
ARC	score_manip	0.35	0.57	0.22
ARC	position	0.38	0.62	0.24
ARC	verbosity	0.44	0.71	0.27
ARC	authority	0.36	0.60	0.24
ARC	few_shot	0.32	0.53	0.21

Table 19: Representative Pareto-optimal panels on the (K_{eff}, $\eta_{\max}$) frontier. All consist of large-tier or large+medium models.

ID	**Models**	K_{eff}	$\eta_{\max}$	**Mean ASR**	**Composition**
P_012	GPT-4.1 / GPT-4o / Qwen72B	1.91	0.08	0.142	all-large
P_034	GPT-4.1 / Llama70B / GPT-4o	2.14	0.09	0.151	all-large
P_087	GPT-4.1 / GPT-4o / Claude-Haiku	1.72	0.07	0.138	large+medium
P_156	GPT-4o / Qwen72B / Llama70B	2.31	0.11	0.163	all-large
P_201	GPT-4.1 / GPT-4.1m / GPT-4o	1.45	0.07	0.131	all-large